Quantum Continual Learning via Fisher EWC and Experience Replay in Variational Quantum Circuits

D. Martín-Pérez[1], F. Rodríguez-Díaz[1], D. Gutiérrez-Avilés[2], A. Troncoso[1], and F. Martínez-Álvarez[1]

[1] Pablo de Olavide University, Seville, Spain
dmarper2@upo.es
[2] University of Seville, Seville, Spain

Abstract. Catastrophic forgetting prevents quantum machine learning models from retaining previously learned tasks when trained sequentially. This paper presents Quantum Transfer Continual Learning (QTCL), a hybrid quantum-classical framework combining a trainable encoder, Variational Quantum Circuits with task-shared and task-specific parameter banks, Fisher Information-based Elastic Weight Consolidation computed via adjoint differentiation, and an experience replay buffer. On Split-MNIST (5 sequential binary tasks, 3 seeds), QTCL achieves average accuracy 0.844 ± 0.012 and forgetting 0.130 ± 0.023, a $+20.8$ pp gain and $3.3\times$ forgetting reduction over the best classical baseline. A noise robustness analysis under the IBM Brisbane calibration noise model shows only a 2.6 % accuracy drop, confirming compatibility with near-term quantum hardware.

Keywords: Quantum Machine Learning · Continual Learning · Variational Quantum Circuits · Elastic Weight Consolidation · Barren Plateaus · NISQ

1 Introduction

In many practical settings, a machine learning model must learn a sequence of tasks over time without access to prior data. This scenario, known as *continual learning*, is hindered by *catastrophic forgetting* [3, 11]: gradient updates for new tasks overwrite parameters encoding past knowledge. Classical anti-forgetting strategies fall into three families: regularization [7, 16], replay [13], and architectural methods. In practice, combining regularization and replay yields the best trade-off [13].

Variational Quantum Circuits (VQCs) [1, 6] are parameterized circuits suitable for machine learning in the Noisy Intermediate-Scale Quantum (NISQ) regime. Their parameter space is endowed with the *Quantum Fisher Information Matrix* (QFIM) [4], a Riemannian metric capturing the curvature of the circuit output with respect to parameters. Applying Fisher-based Elastic Weight Consolidation (EWC) to VQC parameters is therefore geometrically motivated: it protects

the directions of highest curvature in quantum parameter space. However, two problems arise. First, the *barren plateau* phenomenon [8, 10] causes gradients to vanish exponentially with circuit depth, making recovery from parameter drift during sequential training hard. Second, highly expressive ansätze are more prone to barren plateaus [5], limiting practical circuit depth.

This paper proposes QTCL, which addresses these problems by combining: (i) QFIM-based EWC to protect quantum parameters geometrically, and (ii) experience replay to provide gradient anchors preventing parameter drift into barren plateau regions. The contributions are as follows.

QTCL framework. An end-to-end hybrid model with MLP encoder, VQC with task-shared and task-specific parameter banks, QFIM-based EWC, and experience replay, evaluated with statistical rigour (3 seeds, 95 % CI).

Noise robustness. QTCL sustains only a 2.6 % accuracy drop under the IBM Brisbane calibration depolarizing noise model, validating near-term hardware compatibility.

Ablation study. Systematic analysis of EWC strength λ and rehearsal ratio ρ, connecting results to barren plateau theory.

2 Related Work

Continual learning. Kirkpatrick et al. [7] introduced EWC, penalizing parameter changes weighted by the diagonal Fisher Information Matrix (FIM). Zenke et al. [16] proposed Synaptic Intelligence, accumulating parameter importance online. Rebuffi et al. [13] introduced rehearsal-based iCaRL. Parisi et al. [12] survey the field comprehensively.

Variational quantum circuits. Cerezo et al. [1] review variational quantum algorithms. The QFIM gives a natural Riemannian metric on quantum circuit parameter space [4]. Holmes et al. [5] connect ansatz expressibility to gradient magnitudes. McClean et al. [10] characterize the barren plateau problem; Larocca et al. [8] review mitigation strategies. Wang et al. [15] show that hardware noise can induce barren plateaus independently of circuit depth.

Quantum continual learning. Chen [2] studied catastrophic forgetting in VQC classifiers and proposed quantum-adapted regularization. QTCL differs in three ways: it computes the FIM exactly via adjoint differentiation rather than approximating it, combines regularization with experience replay, and uses a hybrid encoder decoupling classical pre-processing from the quantum circuit.

3 Methodology

3.1 Architecture

A two-layer MLP maps raw 784-dim MNIST inputs to a bounded latent vector:

$$\mathbf{z} = \tanh\big(W_2\,\mathrm{LN}(\mathrm{ReLU}(W_1\mathbf{x}))\big) \in [-1,+1]^{n_q},$$

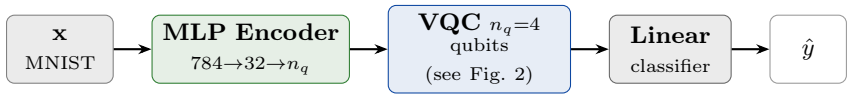


Fig. 1. QTCL architecture. The MLP encoder (green) maps 784-dim inputs to a n_q-dim latent space; the VQC block (blue, detailed in Fig. 2) encodes the latent vector into 4 qubits via R_Y angle encoding, applies shared and task-specific `StronglyEntanglingLayers`, and outputs Pauli-Z expectations to a linear classifier.

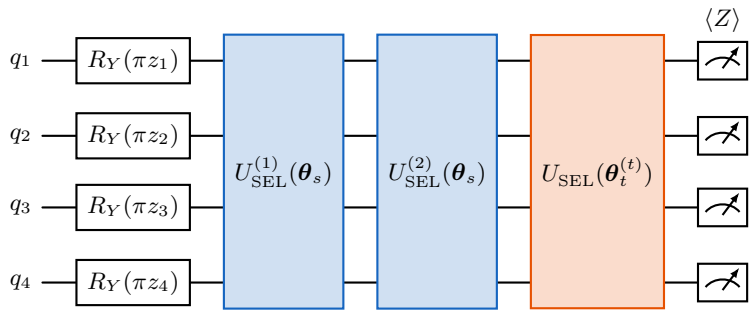


Fig. 2. VQC circuit for $n_q = 4$ qubits. Angle encoding gates $R_Y(\pi z_i)$ map the latent vector to the initial quantum state; two shared `StronglyEntanglingLayers` (blue, $\boldsymbol{\theta}_s$) and one task-specific layer (orange, $\boldsymbol{\theta}^{(t)}_t$) form the variational ansatz; Pauli-Z expectations are read as output features.

where Layer Normalization (LN) prevents encoder drift across tasks. The latent vector is encoded via $R_Y(\pi z_i)$ rotations and processed by:

$$U_{\mathrm{ans}}(\boldsymbol{\theta}_s, \boldsymbol{\theta}^{(t)}_{\mathrm{task}}) = U_{\mathrm{SEL}}(\boldsymbol{\theta}^{(t)}_{\mathrm{task}})\, U_{\mathrm{SEL}}(\boldsymbol{\theta}_s), \tag{1}$$

where $\boldsymbol{\theta}_s$ are *shared* across all tasks, $\boldsymbol{\theta}^{(t)}_{\mathrm{task}}$ are *specific* to task t, and U_{SEL} denotes `StronglyEntanglingLayers` (parameterized $SU(2)$ rotations with all-to-all entanglement). Pauli-Z expectations $\langle Z_i \rangle \in [-1, +1]^{n_q}$ feed a linear classifier. Default: $n_q = 4$, $L_s = 2$ shared layers, $L_t = 1$ task layer (36 quantum parameters total). Figure 1 shows the full pipeline and Figure 2 the VQC circuit structure.

3.2 Fisher EWC for Hybrid Models

The QTCL training loss for task T is:

$$\mathcal{L}_{\mathrm{QTCL}} = \mathcal{L}_{\mathrm{CE}} + \frac{\lambda}{2} \sum_{t<T} \sum_i \hat{F}^{(t)}_i \left(\theta_i - \theta^{(t)*}_i\right)^2, \tag{2}$$

where $\mathcal{L}_{\mathrm{CE}}$ is cross-entropy, $\hat{F}^{(t)}_i$ is the estimated diagonal FIM for parameter i after task t, and $\theta^{(t)*}_i$ is the optimal value of parameter i after task t. The

Algorithm 1 QTCL: Fisher EWC + Experience Replay

Require: Tasks $\{\mathcal{T}_i\}_{i=1}^{T}$; λ_Q, λ_C; ratio ρ; epochs E

Init encoder ϕ, shared $\boldsymbol{\theta}_s$, task params $\boldsymbol{\theta}_{\text{task}}^{(t)}$, classifier W; $\mathcal{B} \leftarrow \emptyset$; $\hat{\mathbf{F}} \leftarrow \emptyset$
for $t = 1$ to T **do**
  Train on $\mathcal{D}_t \cup \mathcal{B}$ with $\mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{EWC}}$ for E epochs
  Compute $\hat{\mathbf{F}}^{(t)}$ (Eq. 3); save $\boldsymbol{\theta}^{*(t)}$
  $\mathcal{B} \leftarrow \mathcal{B} \cup \text{Reservoir}(\mathcal{D}_t, \lfloor \rho |\mathcal{D}_t| \rfloor)$
end for

diagonal FIM is computed via per-sample backward passes:

$$\hat{F}_i^{(t)} = \frac{1}{N} \sum_{n=1}^{N} \left(\frac{\partial \log p(y_n | \mathbf{x}_n, \boldsymbol{\theta})}{\partial \theta_i} \right)^2, \tag{3}$$

where gradients through the VQC use adjoint differentiation. For quantum parameters, the diagonal of $\hat{F}_i$ coincides with the diagonal of the QFIM under the cross-entropy loss when the model is well-calibrated [1], providing a geometric justification: the penalty protects the directions of highest curvature in quantum Hilbert space. Classical (encoder) and quantum (VQC) parameters are treated uniformly with separate strengths $\lambda_C = 500$ and $\lambda_Q = 200$ to account for their different curvature scales ($p_Q = 36$ vs. $p_C \approx 25{,}000$ parameters).

3.3 Training Algorithm

Algorithm 1 describes the full QTCL procedure. After each task, the diagonal FIM is computed on a calibration subset, the parameters are saved as the EWC reference point, and a reservoir sample [14] of past data is added to the replay buffer $\mathcal{B}$, which is mixed with new-task data in all subsequent steps.

3.4 Noise Simulation

To assess noise robustness, a depolarizing noise model is constructed from the calibration data of the IBM Brisbane 127-qubit processor, extracted via the `FakeBrisbane` backend of `qiskit-ibm-runtime`. Single-qubit gates (SX) use per-qubit error rates $p_{1\text{q}} \in \{2.43, 2.43, 3.35, 3.43\} \times 10^{-4}$ (mean $\bar{p}_{1\text{q}} = 2.91 \times 10^{-4}$); two-qubit gates (CX) use $p_{2\text{q}} = 10^{-2}$. Readout errors are not included. The model is applied in PennyLane by switching from `lightning.qubit` (statevector) to `default.mixed` (density matrix), which applies after each gate the depolarizing channel:

$$\mathcal{E}_p(\rho) = (1-p)\,\rho + \frac{p}{3} \sum_{k=1}^{3} \sigma_k \,\rho\, \sigma_k, \tag{4}$$

where $\sigma_k \in \{X, Y, Z\}$ are the Pauli operators and $p \in \{p_{1\text{q}}, p_{2\text{q}}\}$ is the gate-specific error rate. The QTCL model and training loop remain unchanged; only the backend is replaced.

Table 1. CL metrics on Split-MNIST (mean ± std, $n = 3$ seeds, PennyLane backend). ↑: higher is better; ↓: lower is better.

Method	AA↑	BWT↑	FWT↑	F↓
Classical Naive	0.638 ± 0.015	-0.422 ± 0.029	$+0.033 \pm 0.030$	0.422 ± 0.029
Classical EWC	0.636 ± 0.015	-0.426 ± 0.030	$+0.022 \pm 0.019$	0.426 ± 0.030
Quantum Naive	0.599 ± 0.033	-0.462 ± 0.040	$+0.027 \pm 0.031$	0.462 ± 0.040
Quantum EWC	0.617 ± 0.012	-0.454 ± 0.023	$+0.044 \pm 0.036$	0.454 ± 0.023
QTCL (ours)	$\mathbf{0.844 \pm 0.012}$	-0.129 ± 0.024	-0.006 ± 0.063	$\mathbf{0.130 \pm 0.023}$

4 Results

4.1 Setup

Benchmark. Split-MNIST [9, 16] partitions MNIST into 5 sequential binary tasks: $\mathcal{T}_k$: $(2k-2)$ vs $(2k-1)$, $k = 1 \ldots 5$. Each task uses 150 training and 80 test images (balanced).

Metrics. Let $a_{i,j}$ be test accuracy on task j after training through task i. Average Accuracy (AA), Forgetting (F), Backward Transfer (BWT), and Forward Transfer (FWT) are defined as:

$$\mathrm{AA} = \frac{1}{T}\sum_j a_{T,j}, \quad \mathrm{F} = \frac{1}{T-1}\sum_{j<T}(\max_l a_{l,j} - a_{T,j}), \tag{5}$$

$$\mathrm{BWT} = \frac{1}{T-1}\sum_{j<T}(a_{T,j} - a_{j,j}), \quad \mathrm{FWT} = \frac{1}{T-1}\sum_{j>1}(a_{j-1,j} - 0.5), \tag{6}$$

where 0.5 is the random baseline for binary classification.

Methods. Five methods are compared: *Classical Naive* (MLP, no protection); *Classical EWC* (MLP + EWC, $\lambda_C = 500$); *Quantum Naive* (QTCL architecture, no protection); *Quantum EWC* (QTCL + EWC, $\lambda_Q = 200$, no rehearsal); *QTCL* (full system, $\rho = 0.25$). All share the same encoder, Adam ($\eta = 0.002$), batch size 16, and 25 epochs per task.

Infrastructure. All experiments ran on PennyLane `lightning.qubit` (adjoint differentiation) on the Hercules HPC cluster of Universidad Pablo de Olavide (SLURM, 24 cores, 48 GB RAM). Results are means ± std over 3 seeds.

4.2 Main Results

Table 1 reports the CL metrics; Figure 3 shows confidence intervals.

Three patterns stand out from Table 1.

Rehearsal is the critical component. Quantum EWC alone (AA = 0.617) gives no significant improvement over Classical EWC (AA = 0.636). Adding rehearsal yields a +0.227 AA jump, the largest single-component gain. This supports the hypothesis that replay provides the gradient signals needed to keep parameters out of barren plateau regions [10].

Forgetting reduction. QTCL reduces forgetting from 0.426 to 0.130, a 3.3× reduction. Figure 4 shows the 5×5 accuracy matrix $a[i, j]$: QTCL maintains

Table 2. QTCL noise robustness under IBM Brisbane depolarizing noise ($n = 2$ seeds). Δ: absolute change from clean to noisy. The clean condition uses the same seeds as Table 1 subset; minor differences in absolute values reflect seed variance.

Condition	AA	F	BWT
Clean (`lightning.qubit`)	0.866 ± 0.009	0.088 ± 0.006	-0.083 ± 0.002
Noisy (`default.mixed`, FakeBrisbane)	0.840 ± 0.072	0.089 ± 0.048	-0.081 ± 0.050
Δ (noisy − clean)	-0.026	$+0.001$	$+0.002$

near-peak off-diagonal entries throughout sequential training, whereas Classical EWC shows rapid off-diagonal decay after each task transition.

Forward transfer trade-off. While all baselines achieve positive FWT (0.022–0.044), QTCL shows slightly negative FWT (-0.006 ± 0.063). The EWC and replay constraints stabilize shared parameters at the cost of reduced plasticity for zero-shot transfer to unseen tasks. This trade-off is beneficial overall: backward protection (3.3× forgetting reduction) outweighs the marginal loss of forward generalization.

4.3 Ablation Study

Figure 5 shows the ablation results. EWC strength λ_Q peaks in $[200, 500]$ (AA: 0.853 and 0.869, indistinguishable over 2 seeds) and degrades sharply for $\lambda_Q \geq 1000$, when over-constrained parameters prevent effective new-task learning. This optimal range is lower than the classical optimum (~ 500), reflecting the high regularization-per-parameter ratio in the compact VQC space. Rehearsal ratio ρ shows monotonic improvement up to a broad plateau near $\rho = 0.25$–0.40; setting $\rho = 0$ reduces to Quantum EWC (AA = 0.744, consistent with the two-seed ablation setting).

4.4 Noise Robustness

Table 2 shows the noise experiment results. QTCL achieves AA $= 0.840 \pm 0.072$ under the Brisbane noise model, compared to 0.866 ± 0.009 in clean simulation (ΔAA $= -0.026$, -2.6% absolute). The higher clean AA in this experiment relative to Table 1 (0.866 vs. 0.844) reflects seed variance with $n = 2$ seeds. Forgetting is essentially unchanged (ΔF $= +0.001$), confirming that EWC and rehearsal remain effective under depolarizing noise. The dominant noise source is the two-qubit gate error ($p_{2\mathrm{q}} = 10^{-2}$); at the circuit depth used ($L_s = 2$, $L_t = 1$, 4 qubits), the accumulated error remains below the threshold for significant accuracy degradation. The increased standard deviation under noise (0.072 vs. 0.009) reflects shot-to-shot variance from the stochastic depolarizing channel [15]. Figure 6 confirms the results visually.

5 Conclusions

This paper presented QTCL, a hybrid continual learning framework combining QFIM-based EWC (via adjoint differentiation) with experience replay in a VQC with task-shared and task-specific parameter banks. On Split-MNIST (3 seeds, Hercules HPC), QTCL achieves AA $= 0.844 \pm 0.012$ and F $= 0.130 \pm 0.023$, a $+20.8$ pp accuracy gain and $3.3\times$ forgetting reduction over the best classical baseline. Noise analysis under IBM Brisbane calibration data shows only a 2.6 % drop, confirming NISQ compatibility.

Three findings follow. Rehearsal is the critical component: QFIM-based EWC alone gives no benefit over Classical EWC, but the combination yields a $+0.227$ AA gain. The optimal EWC range $\lambda_Q \in [200, 500]$ is lower than for classical models, reflecting the compact VQC parameter space. The anti-forgetting constraints trade a small loss of forward transfer for $3.3\times$ better backward protection.

Limitations include the small dataset scale (150 samples per task), the absence of readout errors in the noise model, and the use of 2 seeds for noise experiments. Future work will extend to Split-CIFAR-10 with barren-plateau mitigation [8], validate on real IBM Quantum hardware with Zero-Noise Extrapolation [15], and study quantum generative rehearsal as a privacy-preserving alternative to data replay.

Acknowledgements

The authors thank the Spanish Ministry of Science and Innovation for support under projects PID2023-146037OB-C21 and PID2023-146037OB-C22, and Pablo de Olavide University for funding the Q-RESILIENCE project. Computations were performed on the Hercules HPC cluster (CITIUS, Universidad Pablo de Olavide). The authors acknowledge the use of IBM Quantum Credits. The views expressed do not reflect the official policy or position of IBM.

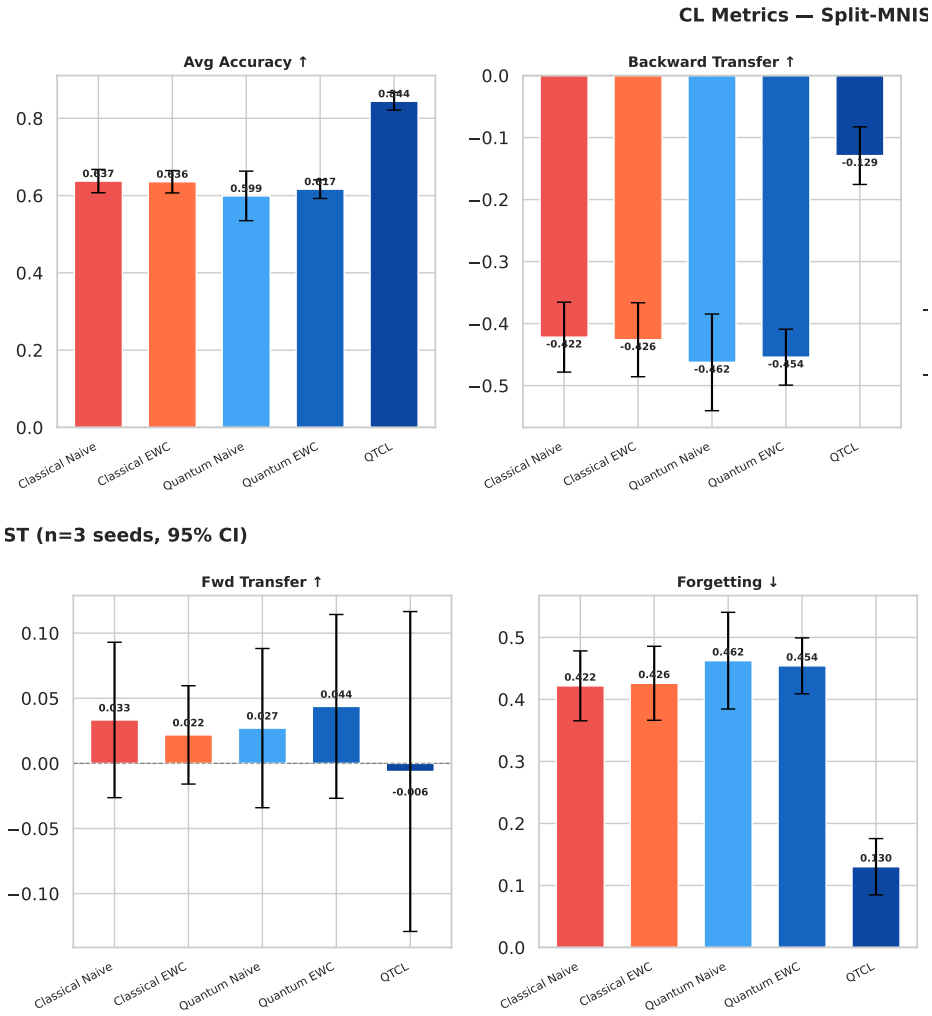
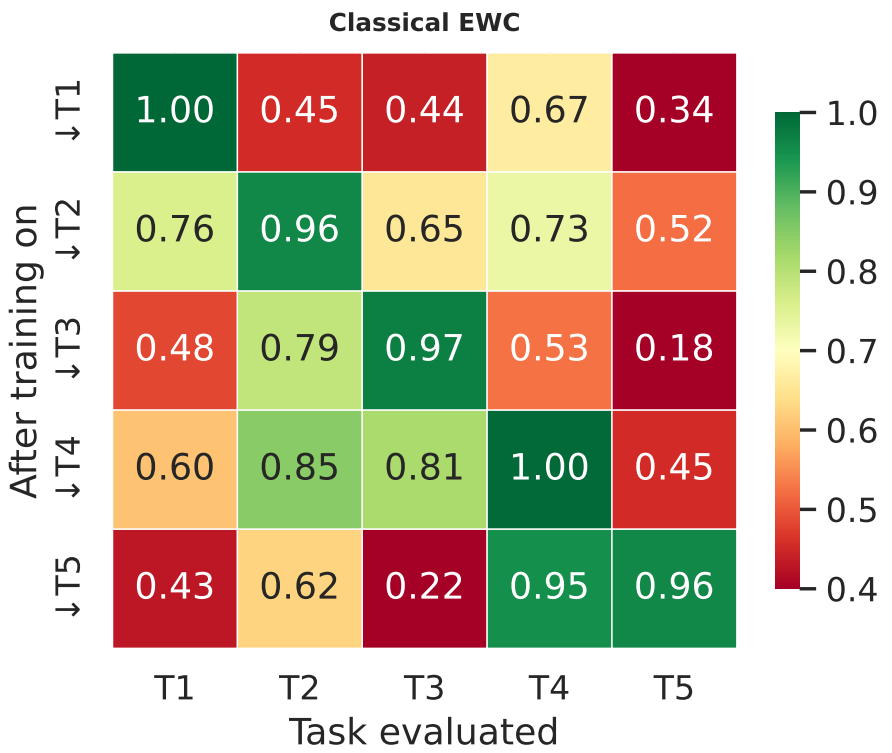
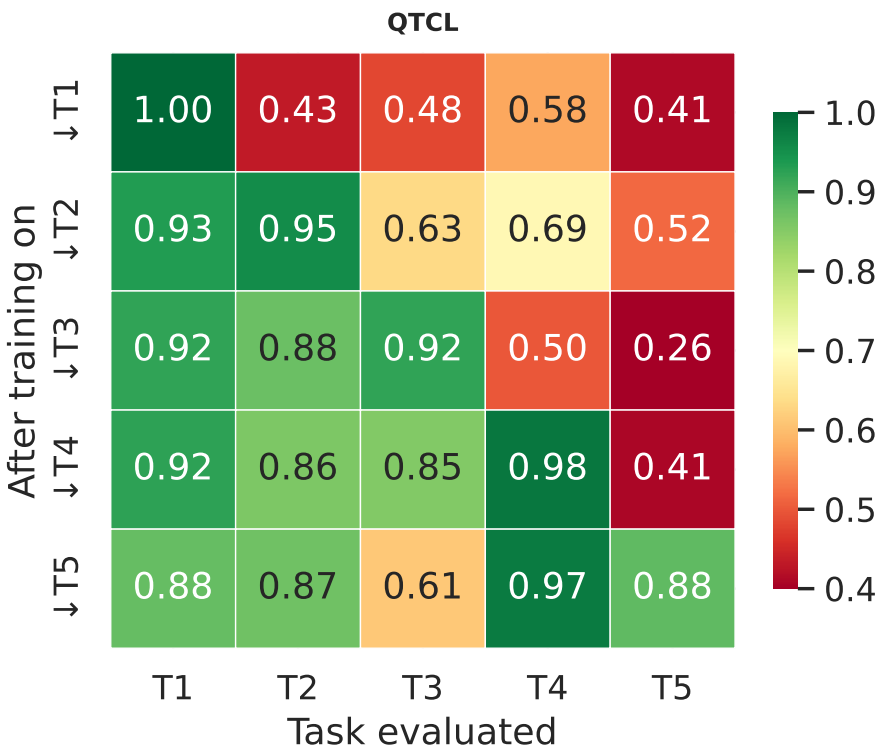


Fig. 3. CL metrics (mean ± 95 % CI, $n = 3$ seeds). QTCL achieves the best AA and F. FWT is best for Quantum EWC (+0.044); QTCL shows slightly negative FWT (–0.006), indicating that the combined EWC and replay constraints on shared parameters reduce zero-shot plasticity to unseen tasks—a trade-off for superior backward protection.

Accuracy Matrix a[i,j] — Split-MNIST





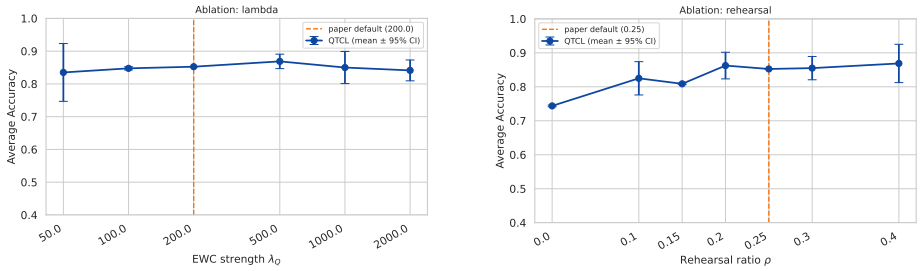


Fig. 5. Left: AA vs. EWC strength λ_Q. Performance peaks in the range $\lambda_Q \in [200, 500]$, reflecting the compact VQC parameter space (36 parameters vs. 25,000 classical). Right: AA vs. rehearsal ratio ρ. A broad plateau is reached near $\rho = 0.25$; $\rho = 0$ degrades to Quantum EWC.

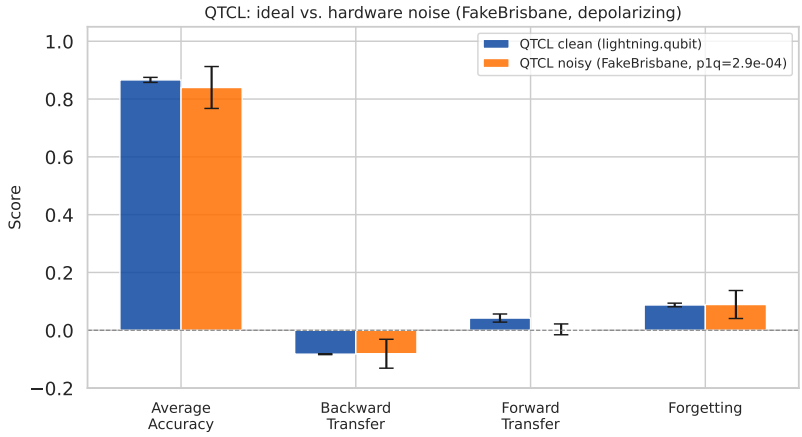


Fig. 6. Clean vs. noisy QTCL under IBM Brisbane depolarizing noise. Average accuracy degrades by only 2.6 %; forgetting is unchanged, confirming that EWC and rehearsal are robust to realistic gate noise.

Bibliography

[1] Cerezo, M., Arrasmith, A., Babbush, R., Benjamin, S.C., Endo, S., Fujii, K., McClean, J.R., Mitarai, K., Yuan, X., Cincio, L., Coles, P.J.: Variational quantum algorithms. Nature Reviews Physics **3**(9), 625–644 (2021)

[2] Chen, S.Y.C.: Quantum continual learning overcoming catastrophic forgetting (2022), arXiv preprint arXiv:2209.10440

[3] Goodfellow, I.J., Mirza, M., Xiao, D., Courville, A., Bengio, Y.: An empirical investigation of catastrophic forgetting in gradient-based neural networks (2013), arXiv preprint arXiv:1312.6211

[4] Haug, T., Kim, M.S.: Optimal training of variational quantum algorithms without barren plateaus (2021), arXiv preprint arXiv:2104.14543

[5] Holmes, Z., Sharma, K., Cerezo, M., Coles, P.J.: Connecting ansatz expressibility to gradient magnitudes and barren plateaus. PRX Quantum **3**(1), 010313 (2022)

[6] Jerbi, S., Fiderer, L.J., Poulsen Nautrup, H., Kübler, J.M., Briegel, H.J., Dunjko, V.: Quantum machine learning beyond kernel methods. Nature Communications **14**, 517 (2023)

[7] Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A.A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., et al.: Overcoming catastrophic forgetting in neural networks. Proceedings of the National Academy of Sciences **114**(13), 3521–3526 (2017)

[8] Larocca, M., Thanasilp, S., Wang, S., Sharma, K., Biamonte, J., Coles, P.J., Cincio, L., McClean, J.R., Holmes, Z., Cerezo, M.: A review of barren plateaus in variational quantum computing. Nature Reviews Physics (2024)

[9] LeCun, Y., Bottou, L., Bengio, Y., Haffner, P.: Gradient-based learning applied to document recognition. Proceedings of the IEEE **86**(11), 2278–2324 (1998)

[10] McClean, J.R., Boixo, S., Smelyanskiy, V.N., Babbush, R., Neven, H.: Barren plateaus in quantum neural network training landscapes. Nature Communications **9**(1), 4812 (2018)

[11] McCloskey, M., Cohen, N.J.: Catastrophic interference in connectionist networks: The sequential learning problem. Psychology of Learning and Motivation **24**, 109–165 (1989)

[12] Parisi, G.I., Kemker, R., Part, J.L., Kanan, C., Wermter, S.: Continual lifelong learning with neural networks: A review. Neural Networks **113**, 54–71 (2019)

[13] Rebuffi, S.A., Kolesnikov, A., Sperl, G., Lampert, C.H.: iCARL: Incremental classifier and representation learning. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 2001–2010. IEEE, Honolulu, HI, USA (2017)

[14] Vitter, J.S.: Random sampling with a reservoir. ACM Transactions on Mathematical Software **11**(1), 37–57 (1985)

[15] Wang, S., Fontana, E., Cerezo, M., Sharma, K., Sone, A., Cincio, L., Coles, P.J.: Noise-induced barren plateaus in variational quantum algorithms. Nature Communications **12**, 6961 (2021)

[16] Zenke, F., Poole, B., Ganguli, S.: Continual learning through synaptic intelligence. In: Proceedings of the 34th International Conference on Machine Learning. pp. 3987–3995. PMLR, Sydney, Australia (2017)